ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Control and Coordination** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Spinal cord reflex centre.
2	**(c)** diabetes	Insulin.
3	**(d)** adrenaline	Fight or flight.
4	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true but not an explanation of neuron direction itself.
5	**(d)** reflex action is faster than brain sensation	Information reaches the brain later.
6	**(d)** iodine	Goitre.
7	**(c)** A is true but R is false.	A true; R false, roots grow downward.
8	**(b)** feel the touch but not move the muscle	Sensory pathway is intact.
9	**(a)** Both A and R are true and R is the correct explanation of A.	The short reflex arc through the spinal cord saves time, which makes the response quick.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; they are different hormones, so R does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Motor neuron or effector (1); Sensory path is intact (1).	1+1
12	Feedback mechanism (1); Level depends on need (1).	1+1
13	Adrenaline released (1); Prepares the body for action (1).	1+1
14	Roots grow into the soil (1); Absorb water and minerals (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Tip makes auxin (1); Auxin goes to the shaded side (1); No tip: no auxin, no bending (1).	1+1+1
16	Motor nerve carries signal to muscle (1); Effector is the muscle (1); Sensory part not needed (1).	1+1+1
17	Adrenaline raises sugar for energy (1); Insulin lowers sugar (1); Opposing action (1).	1+1+1
	OR: Eyes send signals to brain (1); Brain stimulates adrenal glands (1); Adrenaline prepares for flight (1).	1+1+1
18	Chemicals from ovule attract (1); Chemotropism (1); Growth of pollen tube (1).	1+1+1
	OR: Smell receptors are blocked (1); Taste and smell work together (1); Brain combines both (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Reflex action (1). (ii) Motor neuron (1). (iii) Spinal cord (1). (iv) A reflex is quick, involuntary and does not need thinking (1); writing is voluntary, controlled by the cerebrum and slower (1).	1+1+1+2
20	Receptor senses heat (1); Sensory neuron to spinal cord (1); Motor neuron to muscle (1); Hand withdrawn (1); Brain feels pain later and the person screams (1).	1+1+1+1+1
	OR: Insulin: sugar high triggers release (1); Falling sugar stops it (1); Thyroxine: low level stimulates the pituitary (1); Then thyroid makes more (1); Balance is restored (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that a reflex action is controlled by the brain; the reflex arc passes through the spinal cord (the brain is informed later).
- Writing that auxin accumulates on the *lighted* side of the shoot; it accumulates on the shaded side.
- Confusing gland and hormone pairs: adrenaline is from the adrenal gland (not the pituitary), insulin from the pancreas, thyroxine from the thyroid.
- Mixing up the functions of brain parts: cerebellum maintains posture and balance; medulla controls involuntary actions such as blood pressure and vomiting.

- Saying that tropic movements and the movement of Mimosa leaves are the same; tropism is a directional growth, whereas Mimosa's response is a non-directional change in turgor.
- Reading the direction of an impulse wrongly in a neuron; it moves from the dendrite to the cell body, along the axon, and then across the synapse.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Control and Coordination** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** not feel a stimulus	Information does not reach CNS.
2	**(a)** thyroid	Low thyroxine slows metabolism.
3	**(a)** Both A and R are true and R is the correct explanation of A.	Dual function.
4	**(a)** hormones and turgor changes	Chemical coordination.
5	**(b)** insulin	Sugar rises.
6	**(d)** A is false but R is true.	A is false (it is not growth); R is true.
7	**(b)** geotropism	Gravity response.
8	**(c)** adrenaline release	Alertness.
9	**(c)** A is true but R is false.	Insulin lowers the blood sugar level; R is false.
10	**(d)** A is false but R is true.	Adrenaline is secreted by the adrenal glands, so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Time = distance / speed (1); = 0.015 s (1).	1+1
12	Nervous control through brain (1); Oxygen demand rises (1).	1+1
13	Auxin collects on the shaded side (1); More elongation on that side (1).	1+1
14	Goitre (1); Slow growth and low metabolism (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Less insulin (1); Sugar not taken up by cells (1); Sugar appears in urine (1).	1+1+1
16	Longer nerve distance (1); Impulse takes longer to reach the brain (1); Reflex only reduces delay (1).	1+1+1
17	Stem: auxin and cell elongation, slow (1); Leaf: turgor change, fast (1); Growth is permanent; turgor change reversible (1).	1+1+1
	OR: Thyroxine not made (1); Metabolism slows (1); Growth affected (1).	1+1+1
18	Auxin: cell elongation (1); Cytokinin: cell division (1); Work together for growth (1).	1+1+1
	OR: They are proteins, digested in the gut (1); Dosage is controlled by injections (1); Quick action needed (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Insulin (1). (ii) Pancreas (1). (iii) Goitre (1). (iv) Thyroxine, as iodine is required to make it (1); it regulates the metabolism of carbohydrates, proteins and fats for proper growth (1).	1+1+1+2
20	Non-diabetic: sugar rises then falls (1); Insulin rises then falls (1); Diabetic: sugar rises high, stays (1); Low insulin or poor response (1); Treatment through insulin/diet (1).	1+1+1+1+1
	OR: Diabetes (1); Goitre (1); Dwarfism (1); Gigantism (1); Treatments or prevention (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that a reflex action is controlled by the brain; the reflex arc passes through the spinal cord (the brain is informed later).
- Writing that auxin accumulates on the *lighted* side of the shoot; it accumulates on the shaded side.
- Confusing gland and hormone pairs: adrenaline is from the adrenal gland (not the pituitary), insulin from the pancreas, thyroxine from the thyroid.
- Mixing up the functions of brain parts: cerebellum maintains posture and balance; medulla controls involuntary actions such as blood pressure and vomiting.
- Saying that tropic movements and the movement of Mimosa leaves are the same; tropism is a directional growth, whereas Mimosa's response is a non-directional change in turgor.

- Reading the direction of an impulse wrongly in a neuron; it moves from the dendrite to the cell body, along the axon, and then across the synapse.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Control and Coordination** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** stunted growth	No growth hormone.
2	**(a)** Both A and R are true and R is the correct explanation of A.	Result: bending.
3	**(a)** negative geotropism	The shoot grows away from gravity.
4	**(a)** pituitary	Growth hormone.
5	**(a)** Both A and R are true and R is the correct explanation of A.	Fight or flight.
6	**(a)** Both are chemical messengers	Plants have no glands.
7	**(d)** thyroxine	Thyroxine.
8	**(a)** Both A and R are true and R is the correct explanation of A.	Hindbrain function.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Faster elongation on the shaded side bends the shoot towards the light (phototropism).
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; the pituitary is called master gland because it controls other endocrine glands, which R does not state.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Bones need the hormone before growth plates close (1); Growth period is limited (1).	1+1
12	Auxin (1); Promotes growth in shoot, inhibits in root (1).	1+1
13	Insulin is a protein (1); Digested in the stomach (1).	1+1
14	Negative geotropism of shoots (1); Auxin responds to gravity (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	No geotropism (1); Roots grow randomly (1); Shoots depend on light (1).	1+1+1
16	Cerebellum is for balance (1); Precision of movement (1); Posture control (1).	1+1+1
17	Insulates the axon (1); Speeds up conduction (1); Damage slows impulses (1).	1+1+1
	OR: Leaves fold on touch (1); Looks withered to herbivores (1); Turgor change in pulvinus (1).	1+1+1
18	Thyroxine controls metabolism (1); Needed for brain development (1); Iodine is vital in early life (1).	1+1+1
	OR: Animals have neurons (1); Plants lack neurons (1); Plants use chemical signals (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Phototropism (1). (ii) Auxin (1). (iii) On the side away from light (shaded side) (1). (iv) Thigmotropism (response to touch) (1); the part of the tendril in contact with the support grows slowly while the part away from it grows faster, so it coils (1).	1+1+1+2
20	Cerebellum: loss of balance (1); Medulla: involuntary actions fail (1); Pons: breathing disturbances (1); Posture is affected (1); Serious, may be fatal (1).	1+1+1+1+1
	OR: Ear detects (1); Brain interprets (1); Reflex blinks eyes (1); Adrenaline released (1); Heart rate and breathing rise (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that a reflex action is controlled by the brain; the reflex arc passes through the spinal cord (the brain is informed later).
- Writing that auxin accumulates on the *lighted* side of the shoot; it accumulates on the shaded side.
- Confusing gland and hormone pairs: adrenaline is from the adrenal gland (not the pituitary), insulin from the pancreas, thyroxine from the thyroid.

- Mixing up the functions of brain parts: cerebellum maintains posture and balance; medulla controls involuntary actions such as blood pressure and vomiting.
- Saying that tropic movements and the movement of Mimosa leaves are the same; tropism is a directional growth, whereas Mimosa's response is a non-directional change in turgor.
- Reading the direction of an impulse wrongly in a neuron; it moves from the dendrite to the cell body, along the axon, and then across the synapse.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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